

Title	🔗 Scope (Model)	Task	🔗 Scope (SE)	Taxonomy Provided	Mapping Between	🔗 Environment	🔗 Method	Automated Categorization	Notes	🔗 Database	Main Objective
Automated categorization of pre-trained models in software engi PTM		6 macro tasks	Partial SDLC	-	Tag-SETask	Hugging Face	Repository Mining	yes	The paper's main contribution is a semi-automated approach to map pre-trained models (PTMs) from Hugging Face to software engineering (SE) tasks using tags and model metadata.	ACM	This paper introduces an approach to bridge the gap by enabling the automatic classification of PTMs for SE tasks.
How do Pre-Trained Models Support Software Engineering? An PTM		SDLC	SDLC	SDLC based Taxonomy	Model-SETask	Hugging Face	Repository Mining	yes	The paper's main contribution is an automated, SDLC-aligned mapping of PTMs from Hugging Face to 147 SE tasks using model card descriptions, metadata, and arXiv abstracts. It explicitly constructs model-task links.	arxiv	(a) An automated classification framework that accesses the HF API, processes, filters, and classifies PTMs according to the full SDLC. (b) An extensive analysis of SE PTMs, covering their evolution, task distribution, model reuse, and reported attributes. (c) A validated, publicly available dataset of 2,205 SE PTMs.
Large Language Models (LLMs) for Requirements Engineering (LLM		Requirement Engineering	SE Activity	RE Tasks based Taxonomy	Research Paper - SETask	NA	Literature review	no	The paper systematically reviews how LLMs are applied to Requirements Engineering (RE) tasks, categorizes supported RE tasks.		
Large Language Models for Software Engineering: A Systematic LLM		SDLC	SDLC	SDLC based Taxonomy	Research Paper - SETask	NA	Literature review		Comprehensive SLR on LLMs for SE, explicitly categorizes LLMs, SE tasks, and their applications; analyzes capabilities and usage trends across the SDLC.	arxiv	• We are the first to present a comprehensive SLR on 395 papers published between January 2017 and January 2024 that focus on the use of LLM-based solutions to address SE challenges. We conducted a detailed analysis of the selected papers based on publication trends, distribution of publication venues, etc. • We have classified the LLMs utilized for the reported SE tasks and have provided a summary of the usage and trends of different LLM categories within the SE domain. • We describe the reported data processing stages, encompassing data collection, categorization, preprocessing, and representation. • We discuss optimizers used for LLM4SE tasks, including tuning techniques, prevalent prompt engineering techniques, and commonly employed evaluation metrics. • We describe the key applications of LLM4SE encompassing a diverse range of 85 specific SE tasks, grouped into six core SE activities – requirements engineering, software design, software development, software quality assurance, software maintenance, and software management. • We have summarised key challenges that using LLMs encounters within the SE field and have suggested several potential research directions for LLM4SE.
pectives of NLP and Software Engineering: A Survey on Language M Language Models			SDLC	SDLC based Taxonomy	Model-SETask	NA	Literature review		Comprehensive survey of language models (PTMs, LLMs, code-specific models) and their applications across the full software engineering lifecycle, with explicit discussion of tasks, models, and their relationships		
A Survey on Large Language Models for Software Engineering LLM			SDLC	SDLC based Taxonomy	Research Paper - SETask	NA	Literature review		Comprehensive survey of LLMs for SE, explicitly summarizing models, pre-training objectives, and downstream SE tasks across the SDLC. We summarize 62 representative LLMs of Code across three model architectures, 15 pre-training objectives across four categories, and 16 downstream tasks across five categories.		• Survey Methodology. We conduct a detailed analysis of 1009 relevant SE studies empowered with LLMs in terms of publication trends and distribution of venues until mytime. • LLM Perspective. We summarize 62 representative LLMs of Code for the SE community according to different aspects, such as model architectures, pre-training objectives, downstream tasks, and open science. • SE Perspective. We explore the typical application of leveraging the advance of recent LLMs to automate the SE research, involving 947 relevant studies for 112 code-related tasks across five SE phases, i.e., software requirements and design, software development, software testing, software maintenance and software management. • Integration Perspective. We discuss some crucial aspects when LLMs are integrated into the SE field, such as evaluation, benchmarking, security and reliability, and domain tuning. • Outlook and challenges. We pinpoint open research challenges and provide several practical guidelines on applying LLMs for future SE studies.
Software Development Life Cycle Perspective: A Survey of B LLM			SDLC	SDLC based Taxonomy	Benchmark - SETask	NA	Literature review		The paper systematically surveys benchmarks for CodeLLMs and agents across the full SDLC, categorizing benchmarks by SDLC phase and task, and analyzing their coverage and limitations. It provides a structured taxonomy and mapping of benchmarks to SE tasks, directly addressing model-task relationships.	arxiv	• We propose a tiered analysis framework with both general and phase-specific dimensions, and use it to assess how well existing benchmarks align with real-world software engineering needs. • We identify key deficiencies in existing benchmarks and, based on these findings, outline targeted future directions to align benchmarks with realistic engineering scenarios. • We curate and open-source a standardized knowledge base with detailed metadata for all 178 benchmarks, providing a foundational resource for future research on CodeLLMs and agents.
Software Testing With Large Language Models: Survey, Landsc LLM		Software Testing	SE Activity	Software Testing Tasks based Taxonomy	Research Paper - SETask	NA	Literature review		Comprehensive review of LLM use in software testing, analyzing tasks, models, and practices across 102 studies.		- This paper makes the following contributions: • We thoroughly analyze 102 relevant studies that used LLMs for software testing, regarding publication trends, distribution of publication venues, etc. • We conduct a comprehensive analysis from the perspective of software testing to understand the distribution of software testing tasks with LLM and present a thorough discussion about how these tasks are solved with LLM. • We conduct a comprehensive analysis from the perspective of LLMs, and uncover the commonly-used LLMs, the types of prompt engineering, input of the LLMs, as well as the accompanied techniques with these LLMs. • We highlight the challenges in existing studies and present potential opportunities for further studies. • We maintain a GitHub website https://github.com/LLM-Testing/LLM4SoftwareTesting that serves as a platform for sharing and hosting the latest publications about software testing with LLM. • We systematically reviewed the state-of-the-art work in the intersection of software engineering and LLM, a highly discussed topic. We manually selected 123 relevant works from many articles in six databases and conducted detailed organization and categorization. (Open source, https://github.com/KevinHeiwa/LLM-SE-Paper-List). • We categorized these tasks manually into seven types based on different software engineering tasks. For each category, we provided application examples of LLM and delved into detailed explanations. This can assist researchers in better identifying and addressing potential challenges when applying LLM to software engineering tasks. • We have comprehensively collected and compiled the performance of LLM in various software engineering tasks. We have provided an exposition and analysis of LLM's performance in these software engineering tasks, as well as the reasons for variations observed among different studies. This can assist developers in optimizing LLM more effectively.
Towards an understanding of large language models in software LLM		Code Related Taks (seven major software engineering tasks)	Partial SDLC	-	Research Paper - SETask	NA	Literature review			ACM	
Leveraging Large Language Models in the Software Development Lifecycle: Opportunities and Challenges	LLM	LLM capabilities	SDLC	SDLC based Taxonomy	Model Capability - SETask	NA	Literature review		The paper systematically reviews LLM applications across the full SDLC, introduces a unified taxonomy mapping LLM capabilities to SE tasks, and synthesizes empirical evidence and risks.		

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